

pharmaniaga

Lidocaine Hydrochloride

100mg/5mL Injection



DESCRIPTION:

A colourless or pale yellow solution.

COMPOSITION:

Each 5 mL contains Lidocaine hydrochloride 100mg. Preservative free.

PHARMACODYNAMICS:

Lidocaine is a local anaesthetic of the amide type. It is used to provide local anaesthesia at various sites in the body and it acts by inhibiting the ionic refluxes required for the initiation and conduction of impulses, thereby stabilising the neuronal membrane. In addition to blocking conduction in nerve axons in the peripheral nervous system, Lidocaine has important effects on the central nervous system and cardiovascular system. After absorption, Lidocaine may cause stimulation of the CNS followed by depression and in the cardiovascular system; it acts primarily on the myocardium where it may produce decreases in electrical excitability, conduction rate and force of contraction.

PHARMACOKINETICS:

Lidocaine is readily absorbed from the gastrointestinal tract, from mucous membranes, and through damaged skin. Absorption through intact skin is poor. It is rapidly absorbed from injection sites including muscle.

After an intravenous dose Lidocaine is rapidly and widely distributed into highly perfused tissues followed

by redistribution into skeletal muscle and adipose tissue. Lidocaine is bound to plasma proteins, including α 1-acid glycoprotein (AAG). The extent of binding is variable but is about 66%.

Plasma protein binding of Lidocaine depends in part on the concentrations of both Lidocaine and AAG. Any alteration in the concentration of AAG can greatly affect plasma concentrations of Lidocaine.

Plasma concentrations decline rapidly after an intravenous dose with an initial half-life of less than 30 minutes; the elimination half life is 1 to 2 hours but may be prolonged if infusions are given for longer than 24 hours or if hepatic blood flow is reduced.

Lidocaine is largely metabolized in the liver and any alteration in liver function or hepatic blood flow can have a significant effect on pharmacokinetics and dosage requirements. First-pass metabolism is extensive and bioavailability is about 35% after oral doses. Metabolism in the liver is rapid and about 90% of a given dose is dealkylated to form monoethylglycinexylidide and glycinexylidide. Both of these metabolites may contribute to the therapeutic and toxic effects of Lidocaine since their half-lives are longer than that of Lidocaine, accumulation, particularly of glycinexylidide, may occur during prolonged infusions. Further metabolism occurs and metabolites are excreted in the urine with less than 10% of unchanged Lidocaine.

Reduced clearance of Lidocaine has been found in patients with heart failure, alcoholic liver disease, or chronic or viral hepatitis. Drugs that alter hepatic blood flow or induce drug-metabolising microsomal enzymes can also effect the clearance of Lidocaine. Renal impairment may affect the clearance of Lidocaine; accumulation of its active metabolites can also occur.

Lidocaine crosses the placental and blood-brain barrier; it is distributed into breast milk.

INDICATION:

Lidocaine is indicated for production of local or regional anaesthesia by nerve block, infiltration injection, caudal or other epidural blocks.

Treatment or prophylaxis of life-threatening ventricular arrhythmias including those associated with myocardial infarction general anaesthesia in patients predisposed to ventricular arrhythmias, digitalis intoxication or following resuscitation from cardiac arrest

RECOMMENDED DOSAGE:

Lidocaine Injection is for one dose in one patient only. Discard any remaining contents.

As a local anaesthetic for infiltration and nerve block

The dosage varies and depends upon the area to be anaesthetised, vascularity of the tissues, number of neuronal segments to be blocked, individual tolerance and the technique of anaesthesia. The lowest dose needed to provide effective anaesthesia should be administered. For specific techniques and procedures, refer to standard textbooks.

It is recommended that the dose of lignocaine at any one time should not exceed 3mg/kg. However, the dose administered must be tailored to the individual patient and procedure, and the maximum doses here quoted should be used as a guide only.

The normal recommended dose of lignocaine for various anaesthetic procedures in an average, healthy 70kg adult patient are as follows:

Procedure	Concentration (%)	Volume (mL)	Dose (mg)
Infiltration	1.0	20	200
PERIPHERAL NERVE BLOCKS			
Paravertebral	1.0	3 - 5	30 - 50
Pudendal (each side)	1.0	10	100
Paracervical	1.0	10	100
SYMPATHETIC NERVE BLOCK			
Cervical (stellate ganglion)	1.0	5	50
Lumbar	1.0	10	100
EPIDURAL BLOCKS (2 -3mL per segment)			
Thoracic	1.0	10 - 20	100 - 200
Lumbar - analgesia	1.0	10 - 20	100 - 200
- anaesthesia	2.0	5 - 10	100 - 200
Caudal - obstetric analgesia	1.0	10 - 20	100 - 200

The doses suggested above are only a guide. Toxic levels vary widely between patients so doses should be individualised and blood levels should be monitored.

Epidural injections should be administered slowly with frequent aspirations. Subarachnoid and intravascular injections are two of the most serious complications of this technique. If blood or spinal fluid become manifest during aspiration, the needle must be withdrawn and relocated.

The technique of epidural anaesthesia should only be attempted by physicians skilled in this area and readiness for emergencies must be ensured. During spinal anaesthesia the positioning of the patient is very important and the patient's pulse and blood pressure should be monitored. During thoracic, lumbar and caudal epidural anaesthesia, a marked fall in blood pressure and/or intercostal paralysis may be seen, possibly due to the use of excessive doses, improper positioning of the patient or accidental disposition of the anaesthetic within the subarachnoid space. Hypotension and bradycardia may occur as a result of sympathetic blockade.

For continuous epidural or caudal anaesthesia and paracervical block for obstetric analgesia the maximum dose should not be repeated at intervals of less than 1.5 hours.

Adults:

The dose should not exceed 200mg. For spinal anaesthesia, the dose should not exceed 100mg.

Children:

For children, a reduced dosage based on body weight or surface area should be used. The dosage should be calculated for each patient individually and modified in accordance with the physician's experience and knowledge of the patient. Early signs of local anaesthetic toxicity may be difficult to detect in cases where the block is given during general anaesthesia.

In order to minimise the possibility of toxic effects, the use of Lignocaine Injection 1 % solution is recommended for most anaesthetic procedures involving paediatric patients.

The dose should not exceed 3mg/kg.

For intravenous use in cardiac arrhythmias

Patients with congestive heart failure or cardiogenic shock may require smaller bolus doses.

Adults:

The usual dose is lignocaine 50 to 100mg administered intravenously under ECG monitoring. The dose may be injected at a rate of approximately 25 to 50mg (2.5 to 5.0mL of the lignocaine 1% solution or 1.25 to 2.5mL of the 2% solution) per minute. A sufficient period of time should be allowed to enable a slow circulation to carry the drug to the site of action. If the initial dose of 50 to 100mg does not produce the desired response, a second dose may be given after five minutes. No more than 200 to 300mg of lignocaine should be administered during a one hour period.

Following a single injection in those patients in whom arrhythmia tends to recur and who are incapable of receiving oral antiarrhythmic therapy, intravenous infusions of lignocaine may be administered at a rate of 1 to 4mg/minute (20 to 50 mcg/kg/minute). Intravenous infusions must be given under ECG monitoring to avoid potential overdosage and toxicity. The infusion should be terminated as soon as the patient's cardiac rhythm appears to be stable or at the earliest signs of toxicity. It should rarely be necessary to continue the infusion beyond 24 hours. As soon as possible, patients should be changed to an oral antiarrhythmic agent for maintenance therapy.

Paediatrics:

For children, a reduced dose based on body weight or surface area should be used. It is recommended that the 1% solution be used to minimise the possibility of toxic effects. Experience with lignocaine is limited. A suggested paediatric dose is a loading dose of 0.5 to 1mg/kg repeated if necessary up to 3-5mg/kg, followed by continuous infusions of 10 to 50

micrograms/kg/min.

Geriatrics:

A reduction in dosage may be necessary for elderly patients, particularly those with compromised cardiovascular and/or hepatic function.

Impaired hepatic function:

Although lignocaine is metabolised by the liver, dosage reduction for local anaesthesia is probably not warranted. However, caution should be exercised with repeated doses or prolonged infusion.

Impaired renal function:

Impairment of renal function is unlikely to affect lignocaine clearance in the short term (up to 24 hours). However, toxicity due to accumulation may develop with prolonged or repeated administration.

CONTRAINDICATIONS

- Known hypersensitivity to local anaesthetics of the amide type.
- Inflammation or sepsis at the proposed site of injection and in the presence of septicaemia.
- Patients with myasthenia gravis, severe shock or impaired cardiac conduction.
- Epidural or spinal anaesthesia in patients with -uncorrected hypotension or -coagulation disorders or receiving anticoagulants or -serious diseases of the central nervous system or spinal cord such as meningitis, spinal fluid block, cranial or spinal haemorrhage, tumours, poliomyelitis, syphilis, tuberculosis or metastatic lesions of the spinal cord.
- Antiarrhythmic use in patients with -supraventricular arrhythmia or -Stokes-Adams Syndrome or severe degrees of sinoatrial, atrioventricular or intraventricular block unless the patient has an artificial pacemaker.
- Lignocaine suppresses ventricular pacemaker activity and may cause ventricular standstill in such patients.
- General contraindications related to epidural anaesthesia, regardless of the local anaesthetic used, should be taken into account.

WARNINGS AND PRECAUTIONS

As with other local anaesthetics, Lidocaine should be used with caution in patients with epilepsy, myasthenia gravis, cardiac conduction disturbances, congestive heart failure, bradycardia, severe shock, impaired respiratory function or impaired renal function with a creatinine clearance of less than 10mL/minute. Lidocaine is metabolised in the liver and it should be used with caution in patients with impaired hepatic function. Lower doses should be used in congestive cardiac failure and following cardiac surgery.

Hypokalaemia, hypoxia and disorders of acid-base balance should be corrected before treatment with intravenous lidocaine begins.

Facilities for resuscitation should be available when administering local anaesthetics.

The effect of local anaesthetics may be reduced if the injection is made into an inflamed or infected area.

Certain local anaesthetic procedures may be associated with serious adverse reactions, regardless of the local anaesthetic drug used.

- Central nerve blocks may cause cardiovascular depression, especially in the presence of hypovolaemia, and therefore epidural anaesthesia should be used with caution in patients with impaired cardiovascular function.
- Epidural anaesthesia may lead to hypotension and bradycardia. This risk can be reduced by preloading the circulation with crystalloidal or colloidal solutions. Hypotension should be treated promptly
- Paracervical block can sometimes cause foetal bradycardia or tachycardia, and careful monitoring of the foetal heart rate is necessary
- Injections in the head and neck regions may be made inadvertently into an artery, causing cerebral symptoms even at low doses.
- Retrobulbar injections may rarely reach the cranial subarachnoid space, causing serious/severe reactions, including cardiovascular collapse, apnoea, convulsions and temporary blindness
- Retro and peribulbar injections of local anaesthetics carry a low risk of persistent ocular motor dysfunction. The primary causes include trauma and/or local toxic effects on muscles and/or nerves.

The severity of such tissue reactions is related to the degree of trauma, the concentration of the local anaesthetic and the duration of exposure of the tissue to the local anaesthetic. For this reason, as with all local anaesthetics, the lowest effective concentration and dose of local anaesthetic should be used.

Lidocaine Injection is not recommended for use in neonates. The optimum serum concentration of lidocaine required to avoid toxicity, such as convulsions and cardiac arrhythmias, in this age group is not known.

DRUG INTERACTIONS

While adrenaline when used in conjunction with Lidocaine might decrease vascular absorption, it greatly increases the danger of ventricular tachycardia and fibrillation if accidentally injected intravenously.

Dopamine and 5 hydroxytryptamine reduce the convulsant threshold to Lidocaine.

Narcotics are probably proconvulsants and this would support the evidence that Lidocaine reduces the seizure threshold to fentanyl in man.

Opioid-antiemetic combination sometimes used for sedation in children could reduce the convulsant threshold to Lidocaine and increase the CNS depressant effect.

Cimetidine and propranolol depress microsomal enzyme activity, thus enhancing lidocaine toxicity during anti-arrhythmic infusions if concomitantly administered with these drugs, requiring a reduction in the dosage of Lidocaine. Ranitidine produces a small reduction in renal clearance. Increase in serum levels of Lidocaine may also occur with anti-viral agents (e.g. amprenavir, atazanavir, darunavir, lopinavir) Hypokalaemia caused by diuretics may antagonize the action of lidocaine if administered concomitantly.

Cardiovascular collapse has been reported following the use of bupivacaine in patients on treatment with verapamil and timolol; Lidocaine is closely related to bupivacaine.

Lidocaine should be used with caution in patients receiving other local anaesthetics or agents related structurally to amide-type local anaesthetics (e.g. anti-arrhythmics, such as mexiletine), since the systemic toxic effects are additive. Specific interaction studies with lidocaine and class III anti-arrhythmic drugs (e.g. amiodarone) have not been performed, but caution is advised.

There may be an increased risk of ventricular arrhythmia in patients treated concurrently with antipsychotics which prolong or may prolong the QT interval (e.g. pimozide, sertindole, olanzapine, quetiapine, zotepine), prenylamine, adrenaline (if accidentally injected intravenously) or 5HT₃ antagonists (e.g. tropisetron, dolasetron).

Concomitant use of quinupristin/dalfopristin may increase lidocaine levels with a subsequent increased risk of ventricular arrhythmias and therefore should be avoided.

There may be an increased risk of enhanced and prolonged neuromuscular blockade in patients treated concurrently with muscle relaxants (e.g. suxamethonium).

Incompatibilities:

Lidocaine caused precipitation of amphotericin, methohexital sodium and sulfadiazine sodium in glucose injection. It is recommended that admixtures of lidocaine and glyceryltrinitrate should be avoided.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

Pregnancy

Although animal studies have revealed no evidence of harm to the foetus, Lidocaine crosses the placenta and should not be administered during early pregnancy unless the benefits are considered to outweigh the risks.

Lidocaine readily crosses the placental barrier after epidural or intravenous administration to the mother. The ratio of umbilical to maternal venous concentration is 0.5 to 0.6. The foetus appears to be capable of metabolising Lidocaine at term. The elimination half life in the newborn of the drug received in utero is about three hours, compared with 100 minutes in the adult.

Elevated Lidocaine levels may persist in the newborn for at least 48 hours after delivery. Foetal bradycardia or tachycardia neonatal bradycardia, hypotonia or respiratory depression may occur.

Lactation

Small amounts of Lidocaine are secreted into breast milk and the possibility of an allergic reaction in the infant, albeit remote, should be borne in mind when using Lidocaine in nursing mothers.

ADVERSE REACTIONS

In common with other local anaesthetics, adverse reactions to lidocaine are rare and are usually the result of raised plasma concentrations due to accidental intravascular injection, excessive dosage or rapid absorption from highly vascular areas, or may result from a hypersensitivity, idiosyncrasy or diminished tolerance on the part of the patient. Systemic toxicity mainly involves the central nervous system and/or the cardiovascular system.

Immune system disorders

Hypersensitivity reactions (allergic or anaphylactoid reactions, anaphylactic shock).

Skin testing for allergy to Lidocaine is not considered to be reliable.

Nervous & Psychiatric disorders

Neurological signs of systemic toxicity include dizziness or light-headedness, nervousness, tremor, circumoral paraesthesia, tongue numbness, drowsiness, convulsions, coma.

Nervous system reactions may be excitatory and or depressant. Signs of CNS stimulation may be brief, or may not occur at all, so that the first signs of toxicity may be confusion and drowsiness, followed by coma and respiratory failure.

Neurological complications of spinal anaesthesia include transient neurological symptoms such as pain of the lower back, buttock and legs. These symptoms usually develop within twenty-four hours of anaesthesia and resolve within a few days. Isolated cases of arachnoiditis or cauda equina syndrome, with persistent paraesthesia, bowel and urinary dysfunction, or lower limb paralysis have been reported following spinal anaesthesia with lidocaine and other similar agents. The majority of cases have been associated with hyperbaric concentrations of lidocaine or prolonged spinal infusion.

Blood and Lymphatic System Disorders

Lidocaine may also result in methaemoglobinaemia.

Eye disorders

Blurred vision, diplopia and transient amaurosis may be signs of lidocaine toxicity.

Bilateral amaurosis may also be a consequence of accidental injection of the optic nerve sheath during ocular procedures. Orbital inflammation and diplopia have been reported following retro- or peribulbar anaesthesia.

Ear and labyrinth disorders

Tinnitus, hyperacusis.

Cardiac and vascular disorders

Cardiovascular reactions are depressant and may manifest as hypotension, bradycardia, myocardial depression, cardiac arrhythmias and possibly cardiac arrest or circulatory collapse.

Hypotension may accompany spinal and epidural anaesthesia. Isolated cases of bradycardia and cardiac arrest have also been reported.

Respiratory, thoracic or mediastinal disorders

Dyspnoea, bronchospasm, respiratory depression, respiratory arrest.

Gastrointestinal

Nausea, vomiting.

Skin & subcutaneous tissue disorders

Rash, urticaria, angioedema, face oedema.

OVERDOSAGE

Symptoms of acute systemic toxicity:

Central nervous system toxicity presents with symptoms of increasing severity. Patients may present initially with circumoral Paraesthesia, numbness of the tongue, light-headedness, hyperacusis and tinnitus. Visual disturbances and muscular tremors or muscle twitching are more serious and precede the onset of generalised convulsions. These signs must not be mistaken for neurotic behaviour. Unconsciousness and grand mal convulsions may usually follow, which may last from a few second to several minutes. Hypoxia and hypercapnia occur rapidly following convulsions due to increase muscular activity, together with the interference with normal respiration and loss of the airway. In severe cases, apnoea may occur. Acidosis increases the toxic effects of local anaesthetics.

Effects on the cardiovascular system may be seen in severe cases. Hypotension, bradycardia, arrhythmia and cardiac arrest may occur as a result of high systemic concentrations, with potentially fatal outcome.

Recovery occurs as a consequence of redistribution of the local anaesthetic drug from the central nervous system, and metabolism and may be rapid unless large amounts of the drug have been injected.

Treatment of acute toxicity:

If signs of acute systemic toxicity appear, injection of the anaesthetic should be stopped immediately.

Treatment will be required if convulsions and CNS depression occurs. The objectives of treatment are to maintain oxygenation, stop the convulsions and support the circulation. A patent airway should be

established and oxygen should be administered, together with assisted ventilation (mask and bag) if necessary.

The circulation should be maintained with infusions of plasma or intravenous fluids. Where further supportive treatment of circulatory depression is required, use of a vasopressor agent may be considered although this involves a risk of CNS excitation. If convulsion do not stop spontaneously in 15-20 seconds, they may be controlled by the intravenous administration of Diazepam or Thiopentone Sodium, bearing in mind that anti-convulsant drugs may also depress respiration and the circulation. Prolonged convulsions may jeopardize the patient's ventilation and oxygenation and early endotracheal intubation should be considered. If cardiac arrest should occur, standard cardiopulmonary resuscitation procedures should be instituted. Continuous optimal oxygenation and ventilation and circulatory support as well as treatment of acidosis are of vital importance.

Dialysis is of negligible value in the treatment of acute overdose with lidocaine.

STORAGE CONDITION

Store below 30°C.

Do not use if solution contains particles, growth, turbidity or if there is any change of appearance.

SHELF-LIFE

2 years from date of manufacture

Discard unused portion.

DOSAGE FORMS AND PACKAGING AVAILABLE

10 x 5ml ampoule (clear)

10 x 5mL vial (clear)

REGISTRATION NO:

MALxxxxxxxx

PRODUCT REGISTRATION HOLDER / MANUFACTURER:

PHARMANIAGA LIFESCIENCE SDN BHD (82685-T)

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